一、循环积分.(广义动量守恒)

拉氏函数一般是广义坐标，广义速度与时间的函数.

若拉氏函数不显含某一广义坐标 q_j 时，q_j 称循环坐标或可遗坐标.

与此循环坐标对应的存在着运动积分(守恒量)：

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right)-\frac{\partial L}{\partial q_j}=0.$$

$$\frac{\partial L}{\partial q_j}=0$$

$$\Rightarrow \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right)=0$$

$$\Rightarrow \quad \frac{\partial L}{\partial \dot{q}_j}\equiv p_j=const.$$

该常量由初始条件决定，称广义动量积分.

注：广义动量积分无可加性，一般也不对应某具体质点的动量.

二、能量积分（广义能量守恒）

若一力学体系具有**时间平移对称性**，则其拉格朗日函数不显含时间 t，此时系统存在另一运动积分：（广义能量守恒）：

$\frac{\partial L}{\partial t}=0$

$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j}-\frac{\partial L}{\partial q_j}=0 \Rightarrow \frac{\partial L}{\partial q_j}=\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_j}$

$$\frac{dL}{dt}=\sum_{j=1}^{s}\frac{\partial L}{\partial q_j}\frac{dq_j}{dt}+\sum_{j=1}^{s}\frac{\partial L}{\partial \dot{q}_j}\frac{d\dot{q}_j}{dt}+\frac{\partial L}{\partial t}$$

$$=\sum_{j=1}^{s}\dot{q}_j\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right)+\sum_{j=1}^{s}\frac{\partial L}{\partial \dot{q}_j}\frac{d\dot{q}_j}{dt}=\frac{d}{dt}\left(\sum_{j=1}^{s}\frac{\partial L}{\partial \dot{q}_j}\dot{q}_j\right)$$

$$\Longleftrightarrow \quad \frac{d}{dt}\left(\sum_{j=1}^{s}\frac{\partial L}{\partial \dot{q}_j}\dot{q}_j-L\right)=0.$$

$$\Longrightarrow \sum_{i=1}^{s}\left(\frac{\partial L}{\partial \dot{q}_j}\dot{q}_j\right)-L=\sum_{j=1}^{s}(p_j\dot{q}_j)-L\equiv H=const.$$

哈密顿函数.

这里，H 作为一个守恒量出现，它的物理意义是什么？ **能量**

对 $\vec{r}_i=\vec{r}_i(q_1,q_2,\cdots,q_s,t)$

$$\dot{\vec{r}}_i=\sum_{\alpha=1}^{s}\frac{\partial \vec{r}_i}{\partial q_\alpha}\dot{q}_\alpha+\frac{\partial \vec{r}_i}{\partial t}=\dot{\vec{r}}_i(q_1,q_2,\cdots,q_s,\dot{q}_1,\dot{q}_2,\cdots,\dot{q}_s,t)$$

$$T=\sum_{i=1}^{n}\frac{1}{2}m_i\dot{\vec{r}}_i^{\,2}=\sum_{i=1}^{n}\frac{1}{2}m_i\left(\sum_{\alpha=1}^{s}\frac{\partial \vec{r}_i}{\partial q_\alpha}\dot{q}_\alpha+\frac{\partial \vec{r}_i}{\partial t}\right)^2=T_2+T_1+T_0$$

$$T_2=\sum_{i=1}^{n}\frac{1}{2}m_i\sum_{\alpha=1}^{s}\frac{\partial \vec{r}_i}{\partial q_\alpha}\dot{q}_\alpha\sum_{\beta=1}^{s}\frac{\partial \vec{r}_i}{\partial q_\beta}\dot{q}_\beta=\sum_{\alpha,\beta=1}^{s}\left(\sum_{i=1}^{n}\frac{1}{2}m_i\frac{\partial \vec{r}_i}{\partial q_\alpha}\frac{\partial \vec{r}_i}{\partial q_\beta}\right)\dot{q}_\alpha\dot{q}_\beta=\sum_{\alpha,\beta=1}^{s}A_{\alpha,\beta}\dot{q}_\alpha\dot{q}_\beta$$

$m=2$

$$T_1=\sum_{i=1}^{n}m_i\left(\sum_{\alpha=1}^{s}\frac{\partial \vec{r}_i}{\partial q_\alpha}\dot{q}_\alpha\right)\frac{\partial \vec{r}_i}{\partial t}=\sum_{\alpha=1}^{s}\left(\sum_{i=1}^{n}m_i\frac{\partial \vec{r}_i}{\partial q_\alpha}\frac{\partial \vec{r}_i}{\partial t}\right)\dot{q}_\alpha=\sum_{\alpha=1}^{s}B_\alpha\dot{q}_\alpha$$

$m=1$

$$T_0=\sum_{i=1}^{n}\frac{1}{2}m_i\left(\frac{\partial \vec{r}_i}{\partial t}\right)^2$$

$m=0$.

T_2, T_1, T_0 分别是广义速度的二次、一次与零次齐次式.

利用齐次函数的欧拉定理

> 齐次函数的欧拉定理：
>
> 若 f 为 x_i 的 m 次齐次函数，则：
>
> $$\sum_i \frac{\partial f}{\partial x_i} x_i = mf$$

$$\sum_{\alpha=1}^{s}\left(\frac{\partial T_2}{\partial \dot{q}_\alpha}\dot{q}_\alpha\right) = 2T_2 \quad \sum_{\alpha=1}^{s}\left(\frac{\partial T_1}{\partial \dot{q}_\alpha}\dot{q}_\alpha\right) = T_1 \quad \sum_{\alpha=1}^{s}\left(\frac{\partial T_0}{\partial \dot{q}_\alpha}\dot{q}_\alpha\right) = 0$$

$$T = 2T_2 + T_1 + 0$$

① 稳定约束

$r_i = r_i(q_1, q_2, \cdots, q_s, t)$

L 不显含 t，坐标变换方程不显含 t，

$$\frac{\partial r_i}{\partial t} = 0 \quad T_1 = T_0 = 0 \quad T = T_2$$

$$H = \sum_{\alpha=1}^{s}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\dot{q}_\alpha\right) - L = \sum_{\alpha=1}^{s}\left(\frac{\partial T_2}{\partial \dot{q}_\alpha}\dot{q}_\alpha\right) - L$$

势能 $V = V(q_1, q_2, \cdots, q_s)$

$\frac{\partial V}{\partial \dot{q}_i} = 0 \Rightarrow \frac{\partial L}{\partial \dot{q}_i} = \frac{\partial (T+V)}{\partial \dot{q}_i} = \frac{\partial T}{\partial \dot{q}_i}$

$$= 2T_2 - L = 2T - (T - V) = T + V = E = \text{const}$$

此时 H 的含意，即系统的机械能.

② 非稳定约束，L 不显含 t，坐标变换方程显含 t.

$$\frac{\partial r_i}{\partial t} \neq 0 \quad T = T_2 + T_1 + T_0$$

$$H = \sum_{\alpha=1}^{s}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\dot{q}_\alpha\right) - L = \sum_{\alpha=1}^{s}\left(\frac{\partial T}{\partial \dot{q}_\alpha}\dot{q}_\alpha\right) - L$$

$$= \sum_{\alpha=1}^{s}\left(\frac{\partial T_2}{\partial \dot{q}_\alpha}\dot{q}_\alpha\right) + \sum_{\alpha=1}^{s}\left(\frac{\partial T_1}{\partial \dot{q}_\alpha}\dot{q}_\alpha\right) + \sum_{\alpha=1}^{s}\left(\frac{\partial T_0}{\partial \dot{q}_\alpha}\dot{q}_\alpha\right) - L$$

$$= 2T_2 + T_1 - L = 2T_2 + T_1 - (T_2 + T_1 + T_0 - V) = T_2 - T_0 + V = \text{const}$$

H 的含义为系统的广义能量（不是机械能）

结论

1. 广义能量积分存在条件是：L不显含时间t.(时间平移不变性).

   具体地说，

   - 零约束(封闭系统)或稳定约束(外场不随时间变化的非封闭系统)下，L的能量积分H就是机械能守恒.
   - 在非稳定约束(外场随时间变化的非封闭系统)下，L的能量积分H就是所谓的广义能量守恒.

2. 在哈密顿力学中，H将进一步推广为哈密顿函数，与拉氏函数L一样都是力学体系的特性函数，具广泛的意义与用途.

3. 守恒量是与对称性相联系的，它超出了经典力学，在场与基本粒子的现代理论中都有着广泛的应用.(诺特定理).

4. 实际上在经典力学中，还有2个特别重要的守恒量，它们与空间均匀性与各向同性相关. 动量守恒、角动量守恒.

由时空变换不变性推导动量守恒与角动量守恒.

① 若一个力学体系整体做一平移，力学性质不变，则：其拉格朗日函数也必须不变.

$$\delta L = \sum_{j=1}^{s} \frac{\partial L}{\partial q_j}\delta q_j + \sum_{j=1}^{s} \frac{\partial L}{\partial \dot{q}_j}\delta \dot{q}_j$$

$$= \sum_{j=1}^{s} \delta q_j \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) + \sum_{j=1}^{s} \delta \dot{q}_j \frac{\partial L}{\partial q_j} = \frac{d}{dt}\left(\sum_{j=1}^{s} \frac{\partial L}{\partial \dot{q}_j}\delta q_j\right)$$

又 $\dot{\vec{r}}_i = \sum_{k=1}^{s} \frac{\partial \vec{r}_i}{\partial q_k}\dot{q}_k + \frac{\partial \vec{r}_i}{\partial t}$; $\Rightarrow$ $\frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_k} = \frac{\partial \vec{r}_i}{\partial q_k}$

$$T=\sum_{i=1}^{n}\frac{1}{2}m_i\dot{\vec{r}}_i^{\,2},\quad \frac{\partial L}{\partial \dot{q}_j}=\frac{\partial T}{\partial \dot{q}_j}=\sum_i\frac{\partial T}{\partial \dot{\vec{r}}_i}\cdot\frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_j}=\sum_i m_i\dot{\vec{r}}_i\frac{\partial \dot{\vec{r}}_i}{\partial \dot{q}_j}$$

$$\delta L=\frac{d}{dt}\left(\sum_i m_i\dot{\vec{r}}_i\cdot\delta\vec{r}_i\right)=\frac{d}{dt}\sum_j\left(\sum_i m_i\dot{\vec{r}}_i\cdot\frac{\partial\vec{r}_i}{\partial q_j}\right)\delta q_j$$

$$=\frac{d}{dt}\left(\sum_i m_i\dot{\vec{r}}_i\cdot\sum_j\frac{\partial\vec{r}_i}{\partial q_j}\delta q_j\right)=\frac{d}{dt}\left(\sum_i m_i\dot{\vec{r}}_i\cdot\delta\vec{r}_i\right)$$

空间均匀性：若系统平移一无穷小位移 $\vec{\varepsilon}$，拉格朗日函数不变.

$$\delta L=\frac{d}{dt}\left(\sum_i m_i\dot{\vec{r}}_i\cdot\delta\vec{r}_i\right)=\frac{d}{dt}\left(\sum_i m_i\dot{\vec{r}}_i\cdot\vec{\varepsilon}\right)=\frac{d}{dt}\left(\sum_i m_i\dot{\vec{r}}_i\right)\cdot\vec{\varepsilon}=0$$

由于 $\vec{\varepsilon}$ 具任意性，$\delta L=0\iff\frac{d}{dt}\sum_i m_i\dot{\vec{r}}_i=0$

即有封闭体系总动量守恒：$\vec{P}=\sum_i m_i\dot{\vec{r}}_i=\text{const}$.

- 当存在外场时，空间均匀性遭破坏，总动量不再守恒.
- 而空间变量是三维的，因此外场势能不依赖于某个方向变量时，该方向的总动量分量仍守恒.

② 类似地，若一个力学体系不存在特别的空间方向，即体系做一个整体转动时，力学性质不变，其拉格朗日函数不变

若系统整体转动一个无穷小角度 $\delta\varphi$，则第 i 个质点，位矢将变化：

$$\delta\vec{r}_i=\delta\vec{\varphi}\times\vec{r}_i$$

$$\delta L=\frac{d}{dt}\left(\sum_i m_i\dot{\vec{r}}_i\cdot\delta\vec{r}_i\right)=\frac{d}{dt}\left[\sum_i m_i\dot{\vec{r}}_i\cdot(\delta\vec{\varphi}\times\vec{r}_i)\right]$$

$$=\delta\vec{\varphi}\,\frac{d}{dt}\left(\sum_i\vec{r}_i\times m\dot{\vec{r}}_i\right)=\delta\vec{\varphi}\cdot\frac{d\vec{L}}{dt}=0.$$

由于 $\delta\vec{\varphi}$ 的任意性，$\delta L=0\iff\frac{d\vec{L}}{dt}=0$.

$$\vec{L}=\sum_i\vec{r}_i\times m\dot{\vec{r}}_i=\text{const}$$

即有封闭体系的总角动量守恒.

- 对非封闭体系，若外场为空间各向同性时，总角动量仍守恒.
- 即使外场不再各向同性，若具旋转对称性，则对于该对称轴的转动，系统拉格朗日函数不变，因此系统相对于该对称轴的角动量守恒.